

Metallurgy of Aluminium

Metallurgy :-

The process in which metals are extracted from their ores is called Metallurgy.

Free metals / Native metals / Noble metals Hg, Ag, Pt, Au these metals are directly extracted in pure form.

Ores of Aluminium

Aluminium is the 3rd most abundant element in the earth crust

(a) Silicate Ore :- (SiO_2 in formula)

Kaolin / China Clay $\rightarrow$ china dish

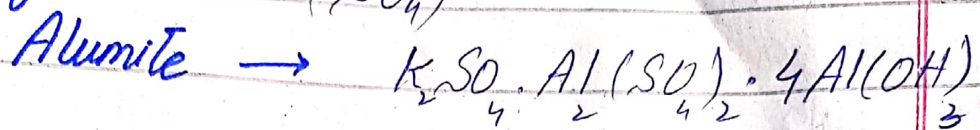
Feldspar $\rightarrow$ Tiles

Mica $\rightarrow$ Electrical insulator (Cut-Out)

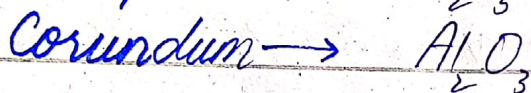
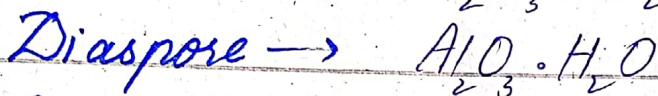
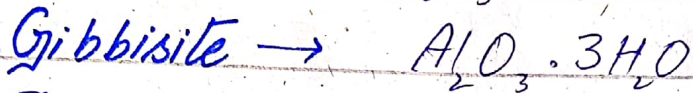
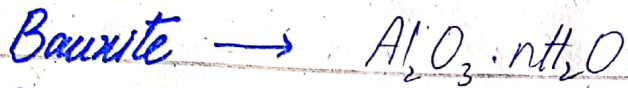
(b) Fluoride Ore : (F)

Cryolite $\rightarrow \text{Na}_3\text{AlF}_6$

(c) Sulfate Ore :- (SO_4)



(d) Oxide Ore : (O)



When corundum extract it
extracted in different colour
used to form Gems. (نگینہ) (نگہ)

Emerald $\rightarrow$ Green (پنا)

Ruby $\rightarrow$ Red (جوتی)

Sapphire $\rightarrow$ Blue (نیلم)

Topaz $\rightarrow$ Yellow (کھراج)

due to different Impurities.

Metallurgy of Al

ORE $\rightarrow$ Bauxite ($\text{Al}_2\text{O}_3 \cdot n\text{H}_2\text{O}$)

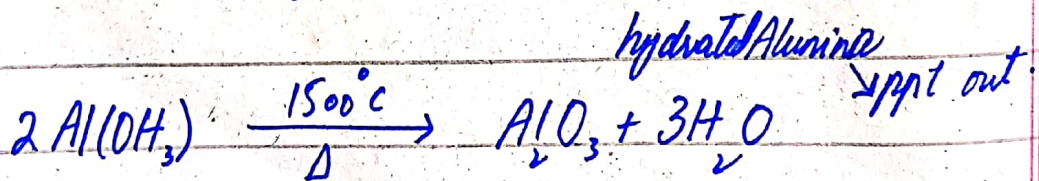
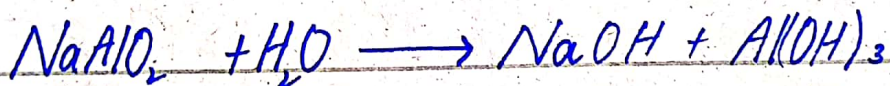
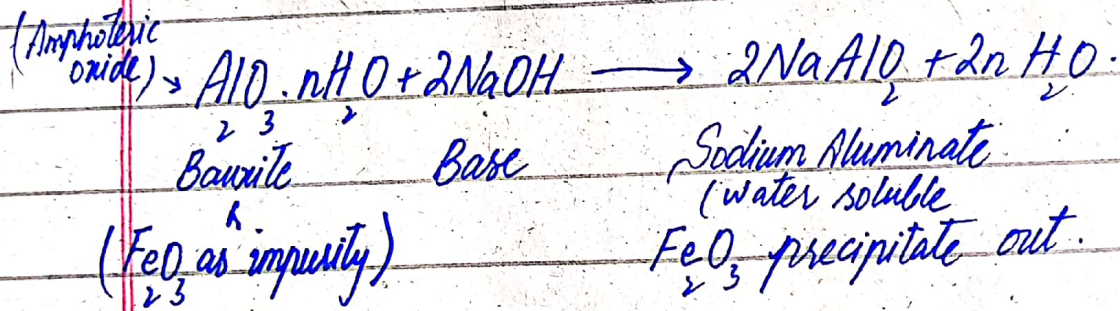
Bauxite are of three types

Red Bauxite (Fe_2O_3 as impurity)
 White " (SiO_2 " "
 Red White " (Both as impurity)

Step 1

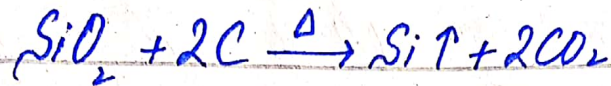
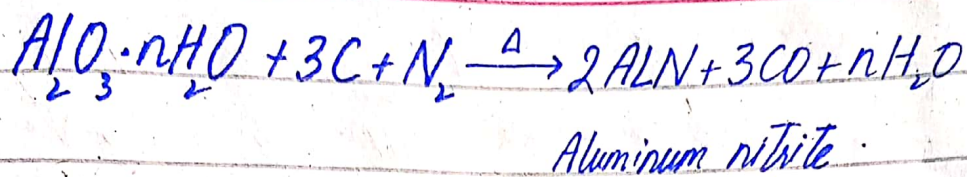
Purification of Bauxite Ore into Al.

1. Bayer's method :- ^{basic oxide}
 If Fe_2O_3 is present as impurity
 then use this method.

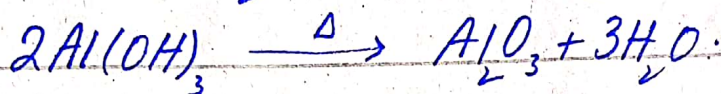
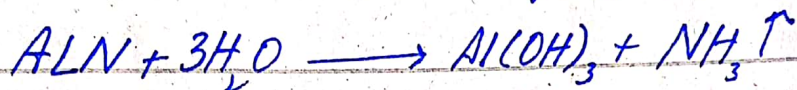


2. Serpek's Method:-

If SiO_2 is present
 as impurity).

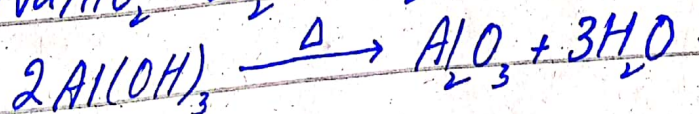
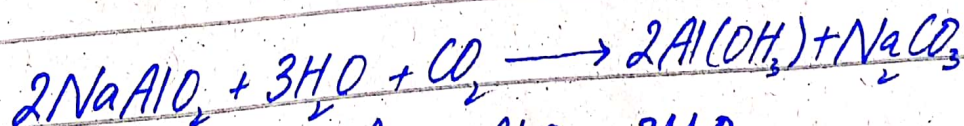
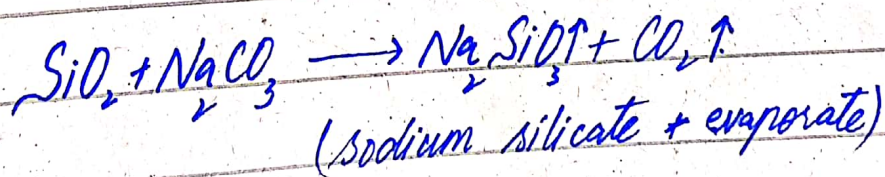
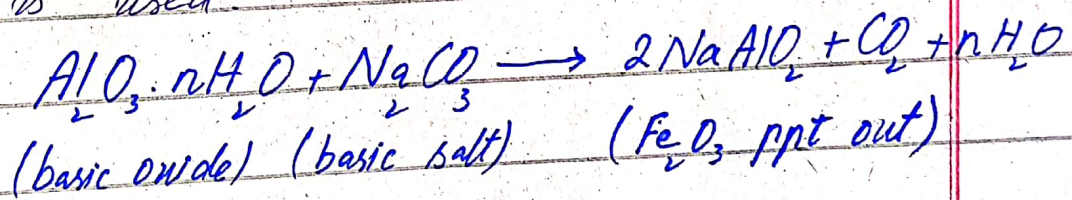


(impurities) hygroscopic in nature



(B) Hall's Method:-

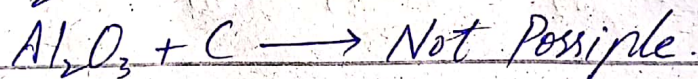
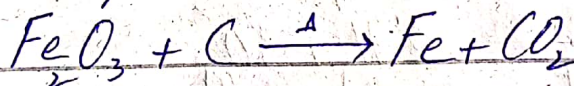
If both impurities are present then this method is used.



Step #2 (Electrolysis of Alumina)

- It is Expensive process due to utilization of Graphite Rods.
- Two methods used in chemistry for reduction of metals.
 - (i) Carbon reduction method.
 - (ii) Electrolysis.

(1) Carbon reduction method can't be used b/c of high reactivity of Al.
It is cheaper method.



HALL - HEROULT'S Electrolytic Cell

Explanation:-

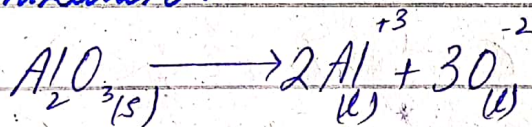
It is basically steel Tank inside Carbon is lining on the wall of tank and Carbon is connected with -ve Terminal. Graphite rods are

dipped in electrolytic mixture filled in the tank. Rods are connected with +ve terminals.

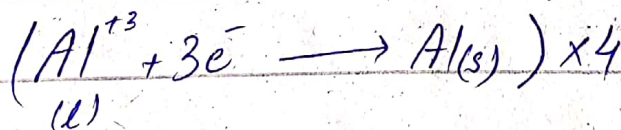
Electrolytic mixture contain Al_2O_3 + Cryolite used for lowering of m.p upto $950^\circ C$ + Fluospar BaF_2 increase fluidity. And Alumina is extracted by passing diaphragm. in molten form 99% pure Al is get.

Rx

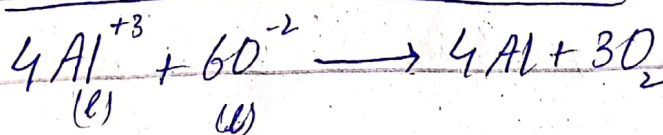
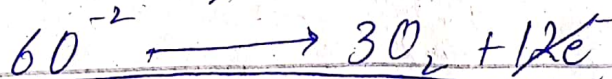
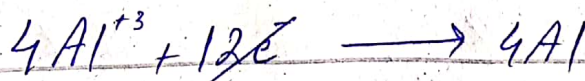
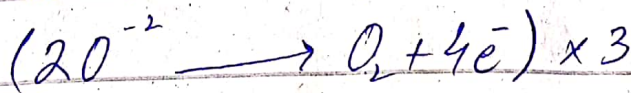
(a) Ionization :-



(b) Rxn Cathode (Reduction)



(c) Rxn Anode (Oxidation)

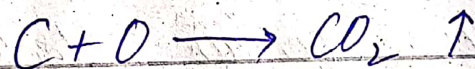


In this method, when we add alumina and current is pass it convert into ion Al^{+3} and O^{-2}

Al^{+3} move towards cathode and O^{-2} move toward anode.

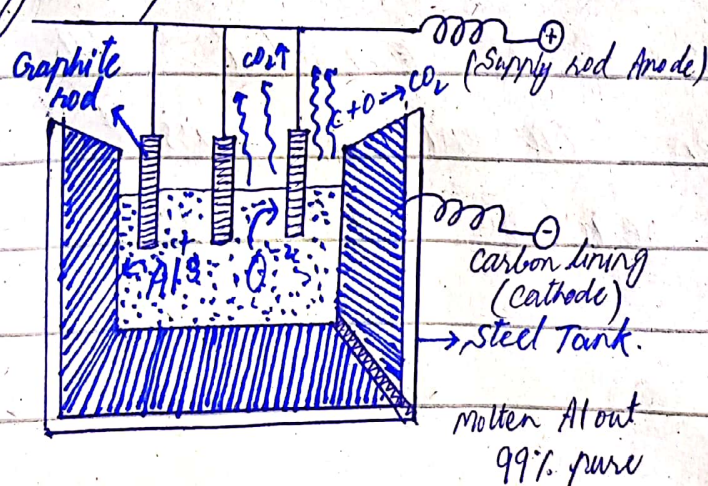
Al^{+3} get $3e^-$ from and convert into Al metal and due to flou spar it extracted from bottom.

O^{-2} move towards C or Graphite rods oxidation is done.



Carbon of rods get oxidation and become reduce slowly.

During process again rods is used in process. Due to this utilization of C or Graphite rod it is very expensive.



Step 3

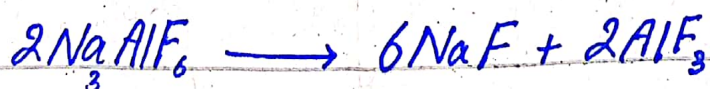
Purification of 99% pure Al into 99.9% pure Al

HOOPE Electrolytic Cell :-

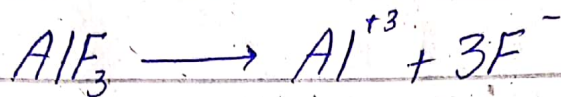
It consists of a tank having three layers of liquid separated by specific gravity. Bottom layer contain 99% Al is present. In Middle layer contain Electrolyte like $(Na_3AlF_6 + BaF_2)$. In this cryolite
Cryolite + Fluorspar

is used in reaction. Fluorspar is used for fluidity. Top layer contain 99.9% pure Al. Top layer connected with cathode and Bottom layer is connected with Anode. When current is passed series of reaction takes place

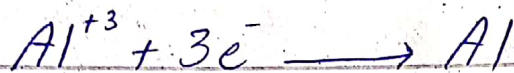
Cell Reactions



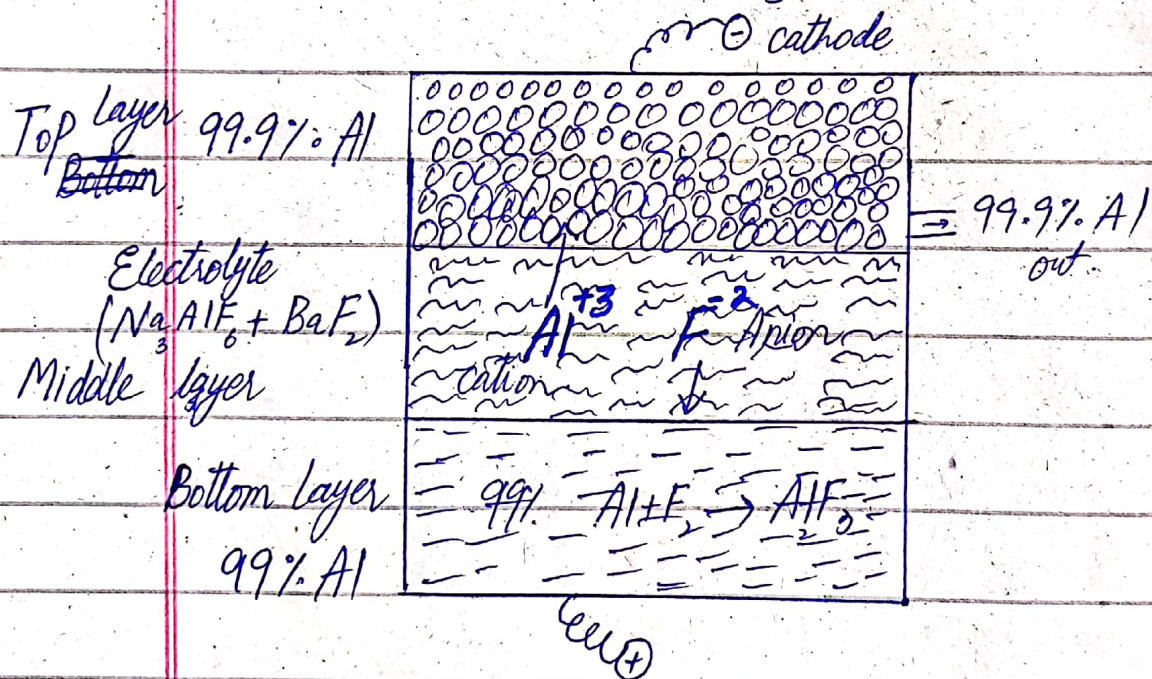
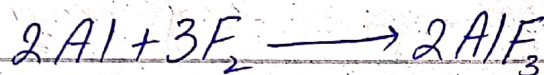
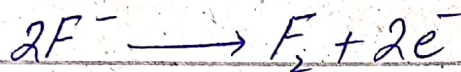
(a) Ionization:-



(b) Rxn at Cathode (Reduction)



(c) Rxn at Anode (Oxidation)



Hall-Héroult Electrolytic Cell